Markov Chains

Definition: The *simple random walk* on $\mathbb{Z}^d$ is given by a walker starting at $(0,\dots,0) \in \mathbb{Z}^d$, then at time t, moves from position

$$x(t) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

to position

$$x(t+1) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

with probability $1/(2d)$ for all $1 \le i \le d$ and every discrete time unit.

Our question is somewhat simple: when, if ever, does the walker return to the origin? This is the substance of Polya's Theorem and its generalization, Varopoulos's Theorem.

Definition: Let X be a finite or countable set. Let $P = (p(x,y))_{x,y\in X}$ be a *stochastic matrix*, a matrix satisfying $p(x,y) \ge 0$ for all $x,y \in X$ and

$$\sum_{y\in X} p(x,y) = 1$$

for all $x \in X$.

We call P the one-step transition probability matrix. The *initial distribution* is a map $\pi\colon X \to \mathbb{R}$ such that $\pi(x) \ge 0$ for all $x \in X$ and $\sum_{x\in X}\pi(x) = 1$. The triple (X,P,π) is called a *Markov chain.*

We may formalize this as follows. Let

$$\Omega = X^{\mathbb{N}}$$

be the space of trajectories. For a fixed $y_0,\dots,y_m \in X$, we let

$$C(y_0,\dots,y_m) = \bigcap_{j=0}^{m} Z_j^{-1}\Big(\big\{y_j\big\}\Big)$$

be the cylinder set with base tuple $(y_0,\dots,y_m) \in X^{m+1}$, where we let $Z_j\colon \Omega \to X$ be the projection map. Define a probability measure $\mathbb{P}_\pi$ by

$$\mathbb{P}_\pi(C(y_0,\dots,y_m)) = \pi(y_0)p(y_0,y_1)\cdots p(y_{m-1}y_m).$$

By the Kolmogorov Extension Theorem, we may uniquely extend $\mathbb{P}_\pi$ to the σ-algebra $\mathcal{B} \subseteq P(\Omega)$ generated by the cylinder sets. We will let $\mathbb{P}$ be this probability measure.

The projection maps Z_j are then X-valued random variables reflecting the position of the walker at time j. The projection maps have the special property known as memorylessness — that is,

$$\begin{aligned} p(x,y) &= \mathbb{P}(Z_{n+1} = y | Z_n = x) \\ &= \mathbb{P}(Z_{n+1} = y | Z_n = x \cap Z_{n-1} = x_{n-1} \cap \cdots \cap Z_0 = x_0). \end{aligned}$$

This is also known as the Markov property.

Conversely, it can be shown that if $(Z_n)_n$ is a sequence of X-valued random variables satisfying the Markov property, then there is a Markov chain (X,P,π) that can be found by taking

$$\pi(x) = \mathbb{P}(Z_0 = x)$$

and determining p as follows: if there exists $n \in \mathbb{N}$ such that $P(Z_n = x) > 0$, set $n(x)$ to be the minimal such value, and else set $n(x) = \infty$; then, set

$$p(x,y) = \begin{cases} P\Big(Z_{n(x)+1} = y | P\Big(Z_{n(x)=x}\Big)\Big) & n(x) < \infty \\ \Phi(y) & \text{else} \end{cases}$$

for all $x, y \in X$, where $\Phi\colon X \to [0,1]$ is a map such that $\sum_{y\in X} \Phi(y) = 1$. If $n(x) < \infty$ for all $x \in X$, then this Markov chain is unique.

Note that Φ is, in a sense, an atomic probability measure.

References

[CSD21] Tullio Ceccherini-Silberstein and Michele D'Adderio. *Topics in Groups and Geometry — Growth, Amenability, and Random Walks*. Springer Monographs in Mathematics. With a foreword by Efim Zelmanov. Springer, Cham, [2021] ©2021, pp. xix+464. ISBN: 978-3-030-88108-5; 978-3-030-88109-2. DOI: 10.1007/978-3-030-88109-2. URL: https://doi.org/10.1007/978-3-030-88109-2.